

Adeel Ahmad

Staff Platform Engineer · Agentic AI and full-stack on AWS · AWS Certified Solutions Architect – Professional

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SUMMARY

Engineer and architect with 19 years of hands-on experience building software and distributed systems, more than ten of them on AWS. I design and build agentic and generative AI systems on Amazon Bedrock AgentCore in Commonwealth Bank's AI Centre of Excellence. Earlier I was an AWS consultant at Cevo, working inside enterprise client teams. Before that I spent nine years as founder and CTO of a software consultancy I grew to 40 people, writing code and selling the work myself.

Most of my recent projects began as ambiguous briefs across several teams, so the first job was working out what was actually needed. I still write a lot of the code myself. I use Claude Code heavily and wrote a plugin for it that holds the agent to an approved plan and to tests that must pass before a task counts as done.

EXPERIENCE

Commonwealth Bank of Australia · Melbourne

Mar 2025 – Present

Staff Platform Engineer (MLOps), AI Centre of Excellence, AI Acceleration squad

- Led architecture and security for CommDataIQ, a data-lineage platform for mainframe modernisation. After narrowing a “map the whole bank” brief to the three tasks data stewards needed most, I built a security-approved prototype with AWS engineers in three weeks, plus an ROI dashboard and a six-month roadmap.
- Took the platform's GenAI agents from prototype to production on Amazon Bedrock AgentCore (Runtime and Memory), with bank-wide rollout planned. Users ask lineage questions in plain English over an Amazon Neptune graph built by parsing Control-M schedules and JCL in Python. Other agents flag broken pipelines across mainframe and cloud systems and suggest fixes that need sign-off before they run.
- Prototyped hybrid RAG (vector search plus cross-encoder reranking, with an LLM explaining each match) in place of a proposed four-week fine-tune for mapping API fields to a 50,000-concept ontology. It worked from about 30 labelled examples where the fine-tune needed 5,000, and data stewards can check the reason behind every mapping.
- Wrote the OAuth2 Proxy sidecar pattern that became the department's reference for service-to-service and agent-to-tool authentication, using workload identity instead of shared secrets. Set up machine-to-machine access to MCP servers with OAuth 2.0 client-credentials tokens checked at a JWT proxy.
- Ran authorised prompt-injection testing of an internal multi-agent assistant; my recorded demo of a trust-boundary gap in its tool use led to guardrail work in the squad. Now writing the spec for an AI agent control tower with three teams.
- Set up the delivery controls for the team's AI services, so prompts are versioned and scored by an LLM judge alongside SAST, DAST and SCA scans, and deploys are staged with rollback. Also scoped the security assessments and pen tests that produced evidence for APRA risk and compliance reviews.
- Raised the squad's DevSecOps maturity score from 1.81 to 2.75 in one cycle (target 2.3) by tying 14 controls to delivery OKRs, and wrote reference docs for CBA's Engineering Skills Framework.
- Mentored engineers and ran department brownbags on authentication patterns and ECS Service Connect security.
- Top of CBA's Cyber Champions leaderboard in February 2026; my team placed third in the bank's GenAI hackathon.

LyreBird Health · Melbourne

Jul 2024 – Oct 2024

Tech Lead, AWS and GenAI (AI medical scribe)

- Built the AWS foundation for patient health data, including a Control Tower multi-account setup, SCPs and least-privilege IAM, KMS key rotation, encryption at rest and in transit, and CI/CD on GitHub Actions using OIDC and Secrets Manager.
- The standard guardrails were distorting sensitive clinical content, so I replaced free-form note generation with two-pass structured extraction in which each statement cites the transcript lines it came from. LLM-as-judge evaluation and prompt versioning let us measure prompt changes on both Amazon Bedrock and OpenAI models.

Cevo (AWS Consulting Partner) · Melbourne

Feb 2022 – Jul 2024

AWS Cloud Consultant, embedded with enterprise clients; multiple performance awards

- Dev lead for government-mandated Consumer Data Right (CDR) REST APIs at a major Australian energy provider, covering the solution architect role early on. Led the domain-API and data-integration teams (8 engineers) and built the AWS data platform behind them; getting the APIs live cleared about \$2.5M of regulatory exposure.
- Fixed a CDR production outage the same day on Amazon Neptune and Gremlin, a stack I learned on the spot after the original team had left. The fix became a bounded-query pattern used across all endpoints.
- On the domain APIs, found accounts with up to 48,000 transactions, well past a 10,000-result limit, and pushed the design to paged queries. Drafted the team's definition of done and reviewed code against each ticket's acceptance criteria.
- In June 2023, pitched the client's CTO on disposable cloud dev environments where an AI agent picks up a Jira story and hands back its plan and a pull request for review. I also proposed a self-hosted coding assistant so source code would stay in-house, and gave the client's dev guild an LLM talk with a local-model demo.

- A meat exporter's CTO wanted a custom SageMaker build. After a week and a half of discovery across three departments, I built a serverless prototype overnight (SES, S3, Step Functions, Textract, Comprehend and Bedrock), and the demo the next morning changed the CTO's mind. The production design came to about \$700 a year for document entry that around 40 people were doing by hand, with no model to train.
- Delivered production retrieval over a 290TB legacy SharePoint estate with Amazon Q Business.

Xoho Tech, Inc · Chicago and Lahore

Jan 2013 – Jan 2022

Founder & CTO, Principal Solutions Architect

- Started the company and grew it to 40 people across two countries. Ran sales myself, from outbound and technical discovery through to proposals and pricing. Delivered 80+ engagements, mostly web and cloud applications, inside client teams at Yale, Harvard, Indiana University/CPB, VIAA (the Flemish archiving institute), the UN and Intel.
- Designed Aviary on AWS for the Yale Fortunoff Archive (4,400+ Holocaust testimonies with strict access control). Built InsureSign, an e-signature product, from scratch on Java/Spring, S3 and SQS; Formstack later acquired it.
- Co-built the open-source Archival Management System with Indiana University/CPB (2.5M records, 1M+ hours of media) and Junkipedia, a misinformation-monitoring platform adopted by the UN. Wrote the engineering playbook.
- Power cuts in Pakistan kept costing developers their work, so I moved development into the browser. Each developer got a VS Code environment built from a project template, shut down when idle and woken on request. It started on one rented server in Europe shared by 12 developers and later moved to Firecracker and Cloud Hypervisor microVMs.

Earlier: PureLogics, Senior Software Engineer then Co-Partner and CTO (2009–2012), where our Facebook app won FBfund. APTLOGIX, Software Developer on e-commerce and C#/.NET (2007–2009).

PROJECTS AND OPEN SOURCE

- **AutoGen notebook (Oct 2023).** Modelled an AWS Professional Services-style team as GPT-4 agents. They scope a customer's architecture brief and check the plan against the Well-Architected Framework, and once a person approves it, an engineer agent drafts code for the specialist agents the plan calls for.
- **agentic-agile** (open source). Claude Code plugin in which a supervisor coordinates nine sub-agent roles, from planning through test-first development, code review and final verification, with each task in its own git worktree. Nothing runs until I approve the plan. Also wrote agent-handoff, a Go tool for passing decisions and open tasks between agent sessions.
- **AgentRC** (agentrc.ai). Working-draft open spec for packaging AI agents. An Agentfile, modelled on a Dockerfile, declares what an agent needs (identity, capabilities, SOPs, policy) and builds into OCI image labels that a platform can enforce with Cedar. Reference CLI in Go with a BuildKit frontend.
- **Agentic Application Studio (Feb 2026).** React prototype, built with Claude, for designing an agent system, from MCP tools and guardrail rules through to synthetic training data and fine-tuning cost estimates on local MLX or Bedrock models.
- **Full-stack TypeScript.** Extended a Next.js, Prisma and PostgreSQL CRM (github.com/adeelahmad/nametag) with Google Workspace integrations and an MCP-based assistant. Built a React/TypeScript exam-prep app on Tauri v2 and SQLite.
- **Acoustic Intelligence** (Hugging Face Space). Gradio app that compares a legacy multi-model audio pipeline with a Gemini-based one, plus a simulated-customer chat.
- **Security and upstream.** Reported unauthorised model selection and system-message tampering in GPTs to OpenAI through Bugcrowd (2024). OpenWrt luci-app-trafficctl fix merged upstream (Aug 2026).

SKILLS

Languages: Python, Java, Go, TypeScript, JavaScript, Ruby, PHP; C#/.NET earlier in my career.

Frontend: React, Next.js, TypeScript, Tauri; Chrome extensions (content scripts, service workers); earlier jQuery and CoffeeScript.

Backend and data: REST API design, microservices, event-driven workflows (SQS, Step Functions), Spring, Rails, Node.js; PostgreSQL, MongoDB, Redis, Amazon Neptune (Gremlin), OpenSearch.

GenAI and agents: Amazon Bedrock, Bedrock AgentCore, Amazon Q Business, MCP, multi-agent orchestration (AutoGen, LangChain); RAG with hybrid retrieval and reranking, embeddings, pgvector, FAISS, Qdrant; prompt engineering and versioning, LLM-as-judge evaluation, guardrails; Claude, Gemini and OpenAI models (OpenAI API beta access since June 2021).

AI-assisted development: Claude Code, OpenCode; sub-agents, hooks and test-first gates; author of Claude Code plugins.

AWS and platforms: Lambda, ECS, EKS, Kubernetes, Docker, S3, RDS, IAM, KMS, Secrets Manager, CloudWatch, X-Ray, Control Tower; CloudFormation, Terraform, Pulumi; GitHub Actions, GitLab CI; Firecracker, Proxmox.

Customer delivery: discovery workshops, working backwards, rapid prototyping, executive presentations, proposals.

Identity and security: OAuth 2.0, OpenID Connect, JWT, SAML, workload identity, Cedar; DevSecOps (SAST, DAST, SCA); prompt-injection testing; APRA, HIPAA, CDR.

CERTIFICATIONS & EDUCATION

AWS Certified Solutions Architect – Professional (valid to Jul 2027; AWS-certified since 2016)

BS Computer Science, Institute of Management Sciences, Lahore (2007–2011)